SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
Chennai- 602105

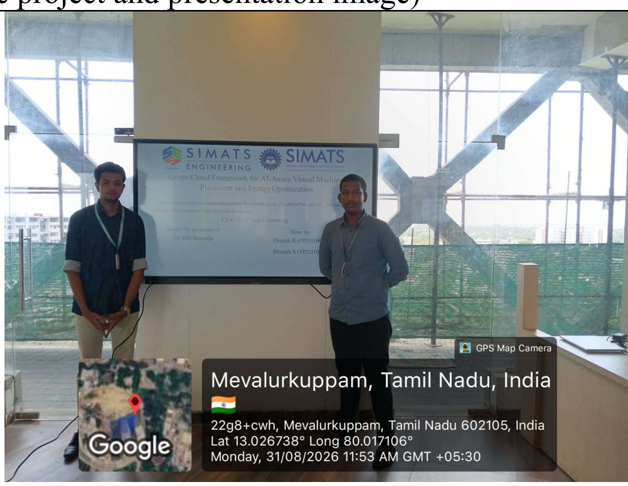
Student Name: Deepak R **Reg. No.:192511008**

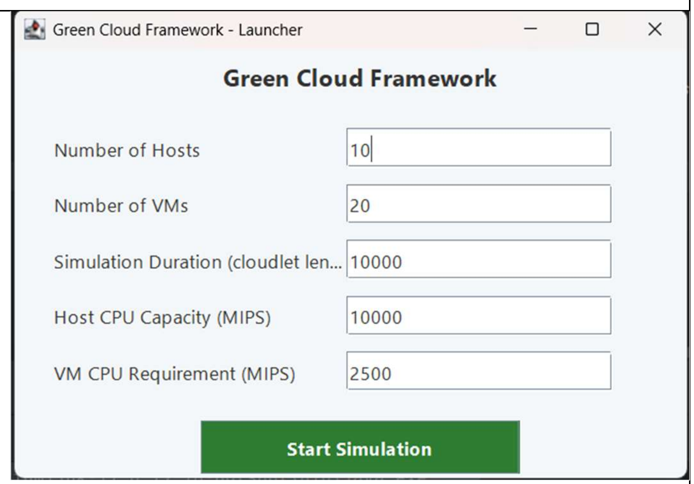
Course Code: CSA1501 **Slot: A**

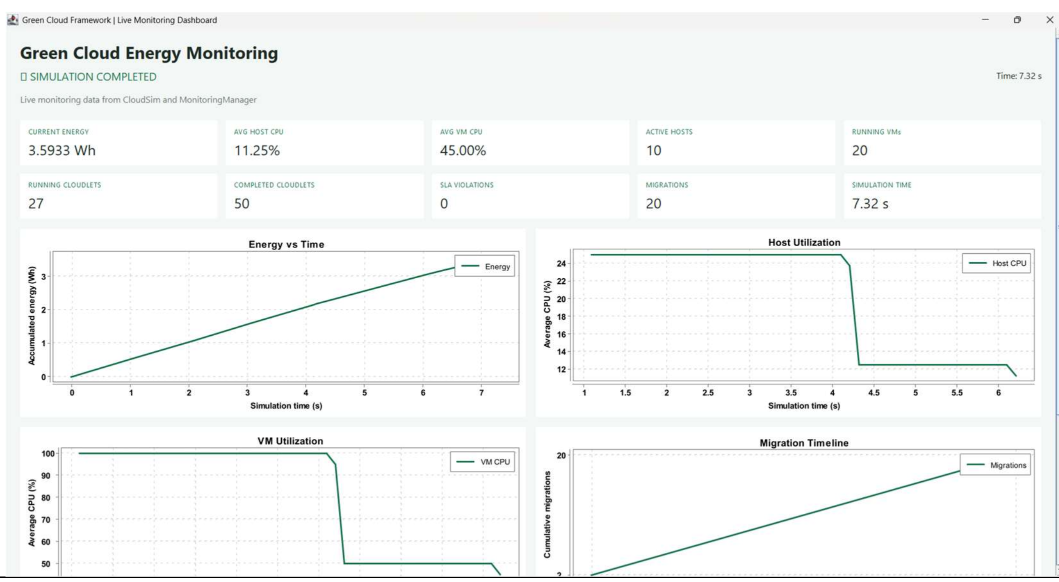
Course Name: CLOUD COMPUTING AND BIG DATA ANALYTICS

Course Faculty: Dr. M. D. Boomija

Project Title: GREEN CLOUD FRAMEWORK FOR AI-AWARE VIRTUAL MACHINE PLACEMENT AND ENERGY OPTIMIZATION
Module Photographs: (3 photographs –Module Photo, Individual student contribution module work in the project and presentation image)







Project Description: (here you write what you did in this project (contribution) including Model Description

The **Green Cloud Framework for AI-Aware Virtual Machine Placement and Energy Optimization** is a software-based cloud simulation system developed using **Java 17, Maven, and CloudSim Plus**. The project simulates cloud data-centre resources such as hosts, virtual machines (VMs), and cloudlets to analyse resource utilization, energy consumption, migrations, and SLA behaviour.

I was responsible for the **overall development and integration of the project**. My work included designing the system architecture, implementing the CloudSim Plus simulation environment, and developing the managers required for datacentre, host, VM, broker, and cloudlet creation. I also implemented the **monitoring subsystem** to collect CPU, memory, VM, cloudlet, energy, migration, and SLA information during simulation.

I developed the **energy model**, using a simplified linear power model with approximately 150 W idle power and 300 W maximum power, and integrated the calculations with the monitoring system. I also worked on the **Live Dashboard**, which displays important KPIs, graphs, and host-level information for understanding simulation performance.

In addition, I implemented **CSV data and report generation**, producing event, host, VM, cloudlet, and simulation-summary reports. I handled the testing and verification of different configurations, analysed the generated results, prepared graphs and tables, and integrated all modules into the final working system. The project was tested using multiple cloud configurations to verify scalability, monitoring, energy reporting, migration tracking, SLA monitoring, and report generation.

Overall, my contribution covered the **complete project lifecycle**, including **planning, architecture, implementation, simulation, monitoring, energy modelling, dashboard development, report generation, testing, result analysis, system integration, and final documentation**.

Student Signature **Guide Signature**